

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 25

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $(15.0 - 2.2) / 15.0 \times 100$ • = 85.3% (accept 85%)	2
01.2	• Added as an anti-knock agent to smooth combustion and reduce engine wear • Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food • Lead is a neurotoxin that bioaccumulates; chronic exposure damages the nervous system / brain development in children	3
01.3	<i>Any 2 from:</i> • Replacement of lead water pipes / lead solder with copper / tin-based solder; phosphoric acid dosing of water supplies forming insoluble lead phosphate • Phasing out of lead-based paint / safer paint removal • Lead-free electrical solder • Restrictions on lead fishing weights / shotgun pellets • Industrial emission controls	2
01.4	• Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe • This prevents lead dissolving into the water	2
01.5	• Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	
02.1	• $38.0 / 0.002$ • = 19 000 (19 000 times / 1.9×10^4)	2
02.2	• Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain)	1

Question	Marking guidance	Marks
02.3	• Persistence: the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain • Liposolubility: the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine • Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level • Herons have long lives so accumulate over time	4
02.4	<i>Any 3 from:</i> • Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification • They persist for decades so the damage continued long after use • Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify • Benefit of preventing malaria deaths outweighs the risk (risk:benefit)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	
03.1	• Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable • Wild species have different ranges of tolerance so cress results cannot be applied directly	2
03.2	• Preliminary study with widely spaced pH values to find where germination occurs • Final range from below the lowest pH giving germination to above the highest, with closer intervals	2
03.3	• Temperature (constant, natural germination temperature) • Light, water volume, number of seeds per dish, seed source • Growth medium (filter paper; soil would alter pH); nutrients not needed for germination • Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure	4
03.4	• Surface water affects fresh mass and drying damages the plants; dry mass requires killing the plants so the same plants cannot be re-measured • Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	

Question	Marking guidance	Marks
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Fossil fuel technologies: secondary / tertiary recovery raising recovery from 20% to 60%; directional drilling and subsea wells; ROVs; fracking for tight oil and gas; tar sands / oil shales; coal gasification / liquefaction; enhanced gas recovery; methane hydrates; CCS (pre / post combustion, storage) • Limits: still finite (discoveries declining since the 1980s); rising energy and financial cost of unconventional sources; environmental impacts (aquifers, water use, tremors, land take, habitat damage); CO₂ and other pollution; CCS unproven at scale and impractical for vehicles • Nuclear: new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash); molten salt reactors; plutonium breeder and thorium reactors increasing fuel use many-fold; fusion as a long-term prospect • Limits: cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium; fusion not commercial • Sustainability criteria: maintaining supply without unacceptable environmental, economic or social impact; comparison with renewables and energy conservation; energy gap argument • Judgement: technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely; role as transition / base-load alongside renewables	9
04.2	• 80-90%	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	

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05.2	• Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	
06.1	• C	1
06.2	• Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions	2
06.3	• Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify • So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
07.1	- The lowest ore purity that can be mined economically with existing technology - Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves)	2
07.2	- X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper - But X is deep: costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) - X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs - X is far from transport infrastructure so transport costs are high - Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall - Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching); both sulfides risk acid mine drainage	6
07.3	<i>Any 2 from:</i> - Higher market value / price of the metal - Improved extraction technology - Lower energy costs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
08.1	C	1
08.2	- Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms - Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas	2
08.3	- HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere - So most do not persist long enough to reach the stratosphere where UV would release their chlorine	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
09.1	- There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) - Most phosphorus compounds have low solubility so little is available in soil water - Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks	3
09.2	- Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil - Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates	2

Question	Marking guidance	Marks
09.3	• Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland	3
09.4	<i>Any 2 from:</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock • Breed crops that absorb phosphate more efficiently • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
10.1	• No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m • Reduction = 2.2 m	2
10.2	• After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream • During dry periods the dam is opened to release water and maintain river levels	2
10.3	• Deforestation / urbanisation reduce infiltration and interception • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods	2

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10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
11.1	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25